

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)			Scale on x -axis 20 °C/division and scale on y -axis 1.0 cm ³ /div (1) All points plotted correctly $\pm < 1$ small square (1) Best fit line judged between 10 – 80 [°C] (1)		3		3	3	3
	(b)			Best fit line extrapolated to 100 [°C] OR 25 [cm ³] (1) At 100 °C the volume of gas will be greater than 25 cm ³ / the syringe limit so it can't be used OR at 90 °C it reaches 25 cm ³ limit so it can't be used (1)		2		2	2	2
	(c)			Volume increases OR separation of molecules increases (1) so density decreases (1)	2			2		2
	(d)			Molecules gain energy OR molecules move more quickly (1) So more <u>frequent</u> collisions (1)	2			2		2
	(e)			$\frac{3.5}{50} = 0.07 \text{ [cm}^3\text{/}^\circ\text{C]} (1)$ $\frac{19.5}{0.07} = 278.6 \text{ [}^\circ\text{C]} (1)$ $10 \text{ }^\circ\text{C} - 278.6 \text{ }^\circ\text{C} = -268.6 \text{ [}^\circ\text{C]} (1)$ Alternative 1: $\frac{50}{3.5} = 14.29 \text{ [}^\circ\text{C/cm}^3\text{]} (1)$ $19.5 \times 14.29 = 278.6 \text{ [}^\circ\text{C]} (1)$ $10 \text{ }^\circ\text{C} - 278.6 \text{ }^\circ\text{C} = -268.6 \text{ [}^\circ\text{C]} (1)$ Alternative for the 3rd mark (if working in Kelvin): $283 \text{ K} - 278.6 \text{ K} = 4.4 \text{ K}$ so $4.4 \text{ K} - 273 \text{ [}^\circ\text{C]} = -268.6 \text{ [}^\circ\text{C]} (1)$ Alternative 2: $\frac{3.5}{50} = 0.07 \text{ [cm}^3\text{/}^\circ\text{C]} (1)$ $\frac{23.0}{0.07} = 328.6 \text{ [}^\circ\text{C]} (1)$ $60 \text{ }^\circ\text{C} - 328.6 \text{ }^\circ\text{C} = -268.6 \text{ [}^\circ\text{C]} (1)$			3	3	3	3

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				Alternative 3: $\frac{50}{3.5} = 14.29 \text{ [}^{\circ}\text{C/cm}^3\text{]} \text{ (1)}$ $23.0 \times 14.29 = 328.6 \text{ [}^{\circ}\text{C]} \text{ (1)}$ $60 \text{ }^{\circ}\text{C} - 328.6 \text{ }^{\circ}\text{C} = -268.6 \text{ [}^{\circ}\text{C]} \text{ (1)}$						
	(f)			Absolute zero	1			1		
				Question 9 total	5	5	3	13	8	12